

When the Secant Numbers Become Periodic

A counterexample, a complete classification,
and a distribution law

Prepared for Vladimir Reshetnikov • 19 September 2026

Abstract

A conjecture recorded in OEIS A000364 asserts that the positive secant numbers, starting at index one, are purely periodic modulo every positive integer m , with a period dividing $\varphi(m)$. It is false: $a_1 = 1$ but $a_{19} \equiv 10 \pmod{27}$, although $\varphi(27) = 18$. We replace the assertion with an exact theorem. Pure periodicity from index one holds precisely when the odd part of m is cube-free. For arbitrary m we determine both the least eventual period and the exact first periodic index. A finite alternating moment formula separates a periodic unit part from a nilpotent part; the latter is exactly $\chi_4(p)p^{2n}a_n$ modulo an odd prime power p^e . A nonvanishing Fermat-quotient coefficient proves sharp period lifting, and a falling-factorial argument handles powers of two.

The classification also yields the natural density of every bounded-preperiod class, a finite-correction Euler product for those densities, and a limiting mean first periodic index. In particular, the original assertion holds for $8/(7\zeta(3)) \approx 95.0751\%$ of moduli. The mean is rigorously enclosed near 1.05402581008837. Exact Python computations, rational numerical certificates, data, and a short standalone counterexample checker accompany the report. The underlying Euler congruences are classical; the purpose is to settle the specific recorded assertion, supply a complete specialized account, and develop its distributional consequences, not to claim that all constituent results are historically new.

The corrected statement. For every $m \geq 1$, the secant numbers are eventually periodic modulo m , with least eventual period dividing $\varphi(m)$. Periodicity already holds for every $n \geq 1$ if and only if no odd prime cube divides m . The smallest exception is $m = 27$.

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1 The problem, its status, and the first counterexample

The positive secant numbers are defined by

$$\sec x = \sum_{n \geq 0} a_n \frac{x^{2n}}{(2n)!}. \quad (1.1)$$

They begin

$$1, 1, 5, 61, 1385, 50521, 2702765, 199360981, \dots$$

and form OEIS A000364. Combinatorially, a_n counts alternating permutations of $2n$ elements with a specified initial direction [1]. We use the signed Euler numbers only as an auxiliary notation:

$$\operatorname{sech} t = \sum_{k \geq 0} E_k \frac{t^k}{k!}, \quad E_{2n} = (-1)^n a_n, \quad E_{2n+1} = 0. \quad (1.2)$$

This distinction matters: the full up/down sequence with generating function $\sec x + \tan x$ is a different sequence, with a different indexing and different periods.

1.1 The assertion selected for investigation

The OEIS entry, as consulted on 19 September 2026, still labels the following assertion as a conjecture and attributes it to Peter Bala, 8 May 2023 [1]:

$$\text{For every } m \geq 1, \quad (a_n \bmod m)_{n \geq 1} \text{ is purely periodic, with a period dividing } \varphi(m). \quad (1.3)$$

Here φ is Euler's totient function. The source's example is $m = 21$, whose least period is indeed 6. Our status claim is deliberately precise: this was a *conjecture still displayed in the cited entry*. It does not establish that no mathematician had previously noticed a counterexample or that its resolution requires new congruence theory.

The distinction between *pure* and *eventual* periodicity is the entire issue. The period-divisibility prediction survives; the claim that there is no transient after index zero does not.

1.2 A finite, independently checkable disproof

From $\sec x \cos x = 1$, coefficient comparison gives

$$a_0 = 1, \quad a_n = \sum_{j=1}^n (-1)^{j+1} \binom{2n}{2j} a_{n-j} \quad (n \geq 1). \quad (1.4)$$

This integer recurrence gives

$$a_{19} = 23489580527043108252017828576198947741.$$

Consequently

$$a_1 \equiv 1 \pmod{27}, \quad a_{19} \equiv 10 \pmod{27}, \quad \varphi(27) = 18. \quad (1.5)$$

A period dividing 18 would force equality at indices 1 and 19. Thus (1.5) disproves (1.3). Appendix A checks this using only the Python standard library and integers of modest size.

Modulo 27, the least eventual period is 18 and the first periodic index is 2. The periodic block at indices 2 through 19 is

$$(5, 7, 8, 4, 11, 1, 14, 25, 17, \\ 22, 20, 19, 23, 16, 26, 13, 2, 10).$$

At index 1, the value is 1, whereas periodic continuation backwards from this block would demand 10. This is a genuine transient, not a mistaken guess about the length of a repeating block.

1.3 What is classical, and what this report establishes

Finite odd-power sums for Euler numbers and their Kummer-type congruences are classical. The sum we use appears explicitly in Sun's paper [4, Eq. (1.1)]; Metsänkylä states the short Kummer congruence, including the Euler factor responsible for the present counterexample [3, p. 161]. Thus a short disproof of the OEIS assertion is already available from established theory.

The modular arithmetic of the *full* up/down sequence was studied by Knuth and Buckholtz [2], then by Ramassamy [5]. A recent preprint of Güleç proves sharp odd-prime-power periods for that full sequence and analyzes its preperiod [6]. The present report uses finite-moment and finite-frequency methods specialized to the even subsequence. All proofs needed below are supplied explicitly. We do not claim to settle the additional preperiod conjectures in that preprint.

The deliverable is a complete resolution of (1.3) as written, together with exact local and global invariants, distribution formulas, and reproducible computations. The density results are deductions proved here; an exhaustive priority claim for them is not made.

2 The complete classification

Definition 2.1. For a fixed modulus $m \geq 1$, let $d(m)$ be the least positive eventual period of $(a_n \bmod m)_{n \geq 0}$. Let $s(m)$ be the least $s \geq 0$ such that

$$a_{n+d(m)} \equiv a_n \pmod{m} \quad (n \geq s).$$

We call $s(m)$ the *first periodic index*, or *preperiod length* when the sequence is indexed from zero. Thus the assertion concerning indices $n \geq 1$ is equivalent to $s(m) \leq 1$.

For an odd prime p , put

$$P_e(p) = p^{e-1} \operatorname{lcm} \left(2, \frac{p-1}{2} \right) = \begin{cases} p^{e-1}(p-1)/2, & p \equiv 1 \pmod{4}, \\ p^{e-1}(p-1), & p \equiv 3 \pmod{4}. \end{cases} \quad (2.1)$$

We write $v_p(u)$ for the exponent of p in a nonzero integer u , and $\chi_4(p) = (-1)^{(p-1)/2}$.

Theorem 2.2 (Exact local and global invariants). *Every modulus has an eventually periodic secant sequence. For an odd prime p and $e \geq 1$,*

$$d(p^e) = P_e(p), \quad (2.2)$$

$$s(p^e) = 1 + \max \{ 0 \leq n < \lceil e/2 \rceil : 2n + v_p(a_n) < e \}. \quad (2.3)$$

The maximum is over a nonempty set. For powers of two,

$$s(2^b) = 0, \quad d(2^b) = \begin{cases} 1, & b = 1, 2, \\ 2^{b-1}, & b \geq 3. \end{cases} \quad (2.4)$$

For $m = 1$, set $s(1) = 0$ and $d(1) = 1$. For every other modulus,

$$s(m) = \max_{p^e \parallel m} s(p^e), \quad d(m) = \operatorname{lcm}_{p^e \parallel m} d(p^e). \quad (2.5)$$

In particular, $d(m) \mid \varphi(m)$ for all $m \geq 1$.

The proof is divided into a finite-transient lemma, the odd-prime argument, the dyadic argument, and the Chinese remainder theorem. Formula (2.3) is finite and effective: it needs only the first $\lceil e/2 \rceil$ secant numbers, no matter how large p^e or its period is.

Corollary 2.3 (Exact replacement of the conjecture). *For every $m \geq 1$, the following are equivalent:*

- (i) $(a_n \bmod m)_{n \geq 1}$ is purely periodic;
- (ii) $s(m) \leq 1$;
- (iii) no odd prime cube divides m .

When these conditions hold, the least period divides $\varphi(m)$. The smallest modulus violating the original assertion is 27.

Deduction from the theorem. If every odd prime exponent is at most two, then for $n \geq 1$ one has $2n \geq e$, so the set in (2.3) consists only of $n = 0$. Thus every odd local preperiod is 1; powers of two cause no transient. Conversely, if $e \geq 3$, then $a_1 = 1$ gives $2 + v_p(a_1) = 2 < e$, so $s(p^e) \geq 2$. The least positive odd prime cube is $3^3 = 27$. $\square$

Table 1: Exact invariants; indexing starts at zero.

m	$s(m)$	$d(m)$	m	$s(m)$	$d(m)$
1	0	1	27	2	18
4	0	1	32	0	16
8	0	4	54	2	18
9	1	6	81	2	54
16	0	8	125	2	50
21	1	6	243	3	162
25	1	10	625	2	250
			3125	2	1250

Notice the two different sources of arithmetic behavior. The period is an explicit multiplicative-order quantity, assembled by least common multiples. The first periodic index depends on divisibility of individual Euler numbers. For example, $s(5^5) = 2$, not the generic upper bound 3, because $a_2 = 5$ supplies an extra factor of 5.

3 A periodic sequence plus a finite transient

The following elementary observation prevents an important logical mistake: observing that one proposed period fails at an early index is not, by itself, proof that *every* possible period fails there.

Lemma 3.1 (Finite-transient separation). *Let A be an abelian group. Suppose a sequence in A has the form $x_n = u_n + b_n$, where u is purely periodic from index zero and $b_n = 0$ for all sufficiently large n . Then:*

- (i) the eventual periods of x are exactly the periods of u ;
- (ii) the least eventual period of x is the least period of u ;
- (iii) if b is nonzero somewhere, the first periodic index of x is $1 + \max\{n : b_n \neq 0\}$; if b is identically zero, it is 0.

Every eventual period starts working at that same first periodic index.

Proof. Let L be any pure period of u . If T is an eventual period of x , then it is an eventual period of u , because $x = u$ on a sufficiently late tail. For any fixed $n \geq 0$, choose k so large that

$n + kL$ lies in that tail. Pure L -periodicity gives

$$u_{n+T} = u_{n+T+kL} = u_{n+kL} = u_n.$$

Thus T is a pure period of u . Conversely, every period of u is clearly an eventual period of x .

Suppose now that T works for x from index s . Since it works for u everywhere, subtraction shows $b_{n+T} = b_n$ for all $n \geq s$. Iterating this equality into the zero tail gives $b_n = 0$ for every $n \geq s$. The converse is immediate. This proves the exact onset statement. $\square$

We will also use the usual divisibility property of periods. If a purely periodic one-sided sequence has periods $P > Q$, then $P - Q$ is a period:

$$u_{n+P-Q} = u_{n+P} = u_n.$$

The middle equality uses the period Q at the nonnegative index $n + P - Q$. The Euclidean algorithm shows that the least period divides every period. The same conclusion holds for eventual periods by passing to a tail. In particular, reducing a sequence modulo a divisor of its modulus forces its least eventual period to divide every eventual period at the larger modulus.

4 Finite moments and the exact odd-prime transient

4.1 An integer identity before reduction

Lemma 4.1 (Finite alternating moment formula). *For every positive odd integer q and every $k \geq 0$,*

$$E_k \equiv \sum_{j=0}^{q-1} (-1)^j (2j+1)^k \pmod{q}. \quad (4.1)$$

Consequently, for $n \geq 0$,

$$a_n \equiv \sum_{j=0}^{q-1} (-1)^j (-(2j+1)^2)^n \pmod{q}. \quad (4.2)$$

At $n = 0$, the convention $x^0 = 1$ includes residue classes $x = 0$.

Proof. Work first in $\mathbb{Q}[[t]]$, not in a ring where factorials might be noninvertible. A finite geometric sum yields

$$\sum_{j=0}^{q-1} (-1)^j e^{(2j+1)t} = \frac{e^t(1 + e^{2qt})}{1 + e^{2t}} = \frac{1 + e^{2qt}}{2 \cosh t}.$$

Comparing the coefficients of $t^k/k!$ gives the exact identity

$$\sum_{j=0}^{q-1} (-1)^j (2j+1)^k = E_k + \frac{1}{2} \sum_{\ell=1}^k \binom{k}{\ell} (2q)^\ell E_{k-\ell}.$$

Every term in the sum on the right is an integer multiple of q . The signed Euler numbers are integers by the coefficient recurrence for $\operatorname{sech} t \cosh t = 1$. Reduction proves (4.1); substituting $k = 2n$ and multiplying by $(-1)^n$ proves (4.2). $\square$

The congruence is known [4, Eq. (1.1)]; the proof above records explicitly why no division by a factorial modulo q is involved.

4.2 Unit and nonunit contributions

Fix an odd prime power $q = p^e$. In $\mathbb{Z}/p^e\mathbb{Z}$, split the last sum into

$$U_n = \sum_{\substack{0 \leq j < p^e \\ p \nmid 2j+1}} (-1)^j (-(2j+1)^2)^n,$$

$$B_n = \sum_{\substack{0 \leq j < p^e \\ p \mid 2j+1}} (-1)^j (-(2j+1)^2)^n.$$

Then $a_n = U_n + B_n$ in this residue ring. Every ratio in U_n is a unit; every ratio in B_n is divisible by p^2 .

Proposition 4.2 (Exact nonunit contribution). *For every $n \geq 0$,*

$$B_n = \chi_4(p)p^{2n}a_n, \quad U_n = (1 - \chi_4(p)p^{2n})a_n \quad \text{in } \mathbb{Z}/p^e\mathbb{Z}. \quad (4.3)$$

Moreover U is purely $P_e(p)$ -periodic and $B_n = 0$ whenever $2n \geq e$.

Proof. The odd multiples of p in the required range are exactly

$$2j+1 = p(2r+1), \quad j = pr + (p-1)/2, \quad 0 \leq r < p^{e-1}.$$

Since p is odd, $(-1)^j = \chi_4(p)(-1)^r$. Therefore

$$B_n = \chi_4(p)p^{2n} \sum_{r=0}^{p^{e-1}-1} (-1)^r (-(2r+1)^2)^n.$$

For $n \geq 1$, Lemma 4.1 makes the inner sum congruent to a_n modulo p^{e-1} . Multiplication by p^{2n} raises the precision to at least p^e . This also covers $e = 1$, where the stated inner congruence is modulo 1. For $n = 0$, both the inner alternating sum and a_0 are exactly 1, so the identity holds directly.

The integer $P_e(p)$ is even and satisfies $\varphi(p^e) \mid 2P_e(p)$. For a unit x modulo p^e , Euler's theorem therefore gives $(-x^2)^{P_e(p)} = 1$. This proves pure periodicity of U . Finally every term in B_n is divisible by p^{2n} . $\square$

Apply Lemma 3.1. The nonzero indices of B are precisely

$$2n + v_p(a_n) < e.$$

They lie below $\lceil e/2 \rceil$ and include $n = 0$. This proves (2.3) and the upper bound $d(p^e) \mid P_e(p)$. Only sharpness of the period remains.

The factor $1 - \chi_4(p)p^{2n}$ is not an arbitrary correction introduced to fit the counterexample. It is exactly what remains after discarding the nonunit bases in the finite moment sum. Its appearance is also the classical Euler factor in Kummer congruences [3].

5 Why the odd-prime-power period is minimal

5.1 Distinct exponential sequences are independent

Lemma 5.1 (Finite-field independence). *Let $\lambda_1, \dots, \lambda_r$ be distinct nonzero elements of a field. If*

$$\sum_{i=1}^r c_i \lambda_i^n = 0$$

holds for r consecutive integer indices $n = N, \dots, N+r-1$, where $N \geq 0$, then every $c_i = 0$.

Proof. The coefficient matrix is a Vandermonde matrix multiplied by the diagonal matrix with entries λ_i^N . Its determinant is

$$\left(\prod_i \lambda_i^N \right) \prod_{i < j} (\lambda_j - \lambda_i),$$

which is nonzero. □

5.2 The period modulo an odd prime

In the moment sum modulo p , pair j with $p - 1 - j$. The corresponding odd residues are negatives of one another, and the two signs $(-1)^j$ agree. Apart from the unique zero residue, the negative squares $\lambda = -(2j + 1)^2$ therefore occur in pairs, each with coefficient $+2$ or -2 . These coefficients are nonzero in $\mathbb{F}_p$.

For positive indices the zero-residue term is absent. Hence an eventual period T gives

$$\sum_{\lambda} c_{\lambda} (\lambda^T - 1) \lambda^n = 0$$

on a tail. By Lemma 5.1, every $\lambda^T = 1$. Taking $\lambda = -1$ forces T to be even. Then $(-x^2)^T = 1$ for all $x \in \mathbb{F}_p^{\times}$ implies $x^{2T} = 1$ for all such x ; since the multiplicative group of a finite field is cyclic,

$$p - 1 \mid 2T.$$

The least positive T satisfying these two conditions is $\text{lcm}(2, (p - 1)/2) = P_1(p)$. The upper bound already proved shows

$$d(p) = P_1(p). \quad (5.1)$$

This is the classical odd-prime period formula recorded in the OEIS with reference to Knuth and Buckholtz [1, 2].

5.3 The lifting step and its nonzero coefficient

Suppose by induction that $d(p^{e-1}) = P_{e-1}(p)$, where $e \geq 2$. Reduction modulo p^{e-1} and the upper bound imply

$$P_{e-1}(p) \mid d(p^e) \mid P_e(p) = p P_{e-1}(p).$$

There are only two possibilities. Set $T = P_{e-1}(p)$; it suffices to prove that T is not an eventual period modulo p^e .

Put

$$\delta = \frac{2P_1(p)}{p-1} \in \{1, 2\}, \quad q_p(x) = \frac{x^{p-1} - 1}{p} \quad (p \nmid x).$$

The integer $q_p(x)$ is the Fermat quotient. Elementary binomial expansion gives

$$x^{2T} = (1 + p q_p(x))^{\delta p^{e-2}} \equiv 1 + \delta p^{e-1} q_p(x) \pmod{p^e}. \quad (5.2)$$

For completeness, the familiar step $(1 + p y)^{p^k} \equiv 1 + p^{k+1} y \pmod{p^{k+2}}$ follows by induction on k , expanding one additional p th power. Raising to $\delta = 1$ or 2 gives (5.2).

For n large enough that the nonunit terms vanish, the moment formula and T even yield

$$\frac{a_{n+T} - a_n}{p^{e-1}} \equiv \delta \sum_{\substack{0 \leq j < p^e \\ p \nmid 2j+1}} (-1)^j q_p(2j+1) (-(2j+1)^2)^n \pmod{p}. \quad (5.3)$$

The quotient on the left is an integer on a sufficiently late tail by the inductive period statement. Equivalently, one may first form the congruence modulo p^e with a factor p^{e-1} on both sides, then reduce the quotient modulo p .

Group the right side by the distinct negative squares modulo p . We compute just one coefficient: the coefficient of $\lambda = -1$, coming from $x \equiv \pm 1 \pmod{p}$. Let $M = p^{e-1}$. The two groups are

$$\begin{array}{c|c|c} x & j & q_p(x) \pmod{p} \\ \hline 1 + 2p\ell & p\ell & -2\ell \\ -1 + 2p(\ell + 1) & p - 1 + p\ell & 2(\ell + 1) \end{array} \quad (0 \leq \ell < M).$$

The quotient formulas follow by expanding x^{p-1} modulo p^2 . Both rows have sign $(-1)^j = (-1)^\ell$. Since M is odd and $p \mid M$,

$$\sum_{\ell=0}^{M-1} (-1)^\ell = 1, \quad \sum_{\ell=0}^{M-1} (-1)^\ell \ell = \frac{M-1}{2}.$$

The first row therefore contributes $-(M-1) \equiv 1 \pmod{p}$, and the second contributes $M+1 \equiv 1 \pmod{p}$. The coefficient of the $(-1)^n$ mode in (5.3) is consequently

$$2\delta \not\equiv 0 \pmod{p}. \quad (5.4)$$

By Lemma 5.1, the normalized defect cannot vanish for all sufficiently large n . Hence T is not an eventual period modulo p^e . The other possibility is forced:

$$d(p^e) = pP_{e-1}(p) = P_e(p).$$

This completes the proof of (2.2).

The important feature is that the obstruction to losing a factor p is the fixed coefficient 2δ . It does not depend on the nonvanishing of a particular large Euler number, and it cannot disappear at an exceptional prime. The use of grouped finite frequencies is closely related to the method used for the full up/down sequence in [6]; here the even subsequence allows the argument to stay in $\mathbb{F}_p$.

6 Powers of two: a self-contained Stern argument

Odd finite moments do not directly address dyadic moduli. We give a short proof of the exact signed-Euler congruence traditionally associated with Stern, then translate it carefully to positive secant numbers. For its classical provenance and other proofs, see Sun [4].

Lemma 6.1 (Dyadic isometry on even Euler indices). *If N, M are distinct nonnegative even integers, then*

$$v_2(E_N - E_M) = v_2(N - M). \quad (6.1)$$

Moreover, for even $N \geq 0$,

$$E_N \equiv 1 - N \pmod{8}. \quad (6.2)$$

Proof. Let $R = \mathbb{Z}_{(2)}$, the ring of rational numbers with odd denominator. The formal series

$$u(t) = \frac{e^{2t} - 1}{2} = \sum_{j \geq 1} \frac{2^{j-1}}{j!} t^j$$

has coefficients in R , since $v_2(j!) \leq j - 1$. Its constant term is zero, so

$$\tanh t = \frac{u(t)}{1 + u(t)} \in R[[t]].$$

Write $\tanh t = \sum_{j \geq 1, j \text{ odd}} b_j t^j$, where $b_j \in R$ and $b_1 = 1$. The exact identity $\operatorname{sech} t = e^t(1 - \tanh t)$ gives

$$E_N = 1 - \sum_{\substack{1 \leq j \leq N \\ j \text{ odd}}} b_j (N)_{\underline{j}}, \quad (X)_{\underline{j}} = X(X-1) \cdots (X-j+1). \quad (6.3)$$

For comparison at N and M , use the same finite summation range up to $\max(N, M)$: terms beyond either nonnegative integer index vanish.

For odd $j \geq 3$, the product $(X)_{\underline{j}}$ has at least two factors with even offsets, namely X and $X - 2$. Telescope the difference of the products evaluated at N and M , changing one factor at a time. Each resulting summand contains one factor $N - M$. After the changed factor is omitted, at least one even-offset factor remains; its value at either even argument is even. Hence

$$(N)_{\underline{j}} - (M)_{\underline{j}} \in 2(N - M)\mathbb{Z} \quad (j \geq 3 \text{ odd}).$$

The $j = 1$ term in (6.3) is $-N$. It follows that

$$E_N - E_M = -(N - M) + 2(N - M)r \quad \text{for some } r \in R.$$

The factor $-1 + 2r$ is a unit in R , proving (6.1).

Finally, for even N and odd $j \geq 3$, the two factors N and $N - 2$ include a multiple of 4 and another multiple of 2. Thus $(N)_{\underline{j}} \in 8\mathbb{Z}$. The same conclusion is automatic for a zero product. In (6.3), all terms except $1 - N$ vanish modulo 8, proving (6.2). $\square$

By $a_n = (-1)^n E_{2n}$,

$$a_n \equiv (-1)^n(1 - 2n) \pmod{8}. \quad (6.4)$$

The repeating residue block is $(1, 1, 5, 5)$, starting at $n = 0$. In particular, every $a_n \equiv 1 \pmod{4}$, so the period modulo 2 and 4 is 1.

For $b \geq 3$, every eventual period must be divisible by 4, since the block modulo 8 has least period 4. For an even shift T , signs at n and $n + T$ agree, and (6.1) gives, at every $n \geq 0$,

$$v_2(a_{n+T} - a_n) = v_2(2T) = 1 + v_2(T).$$

Thus periodicity modulo 2^b is equivalent, among the possible shifts, to $2^{b-1} \mid T$. The least such shift is 2^{b-1} , and it works already at zero. This proves (2.4).

7 Global assembly and a formula for every failure

7.1 Chinese remainders and the exact onset

A congruence modulo m is equivalent to the congruence at every exact prime-power factor of m . An eventual period must therefore be a multiple of each local least period, and the least common multiple is itself an eventual period. This proves the period formula in (2.5).

For the onset, each odd local sequence is a periodic sequence plus a finite transient, as in Lemma 3.1; the dyadic local sequence has no transient. Every local eventual period therefore has the same least valid onset. A global period cannot begin earlier than any local one, while the least common multiple works after the latest local onset. This proves the other formula in (2.5).

Every local period divides $\varphi(p^e)$. Since $\varphi(m) = \prod_{p^e \parallel m} \varphi(p^e)$, every local period divides $\varphi(m)$, and so does their least common multiple. This finishes the proof of Theorem 2.2.

7.2 The defect at a totient shift

Proposition 7.1 (All-index defect). *Let p^e be an odd prime-power divisor of m . For every $n \geq 0$,*

$$a_{n+\varphi(m)} - a_n \equiv -\chi_4(p)p^{2n}a_n \pmod{p^e}. \quad (7.1)$$

In particular,

$$a_{1+\varphi(m)} - 1 \equiv -\chi_4(p)p^2 \pmod{p^e}. \quad (7.2)$$

Proof. The shift $\varphi(m)$ is a multiple of $P_e(p)$, so $U_{n+\varphi(m)} = U_n$. Also $\varphi(m) \geq \varphi(p^e) \geq e$ for odd p , so $B_{n+\varphi(m)} = 0$. Since $a = U + B$, the difference is $-B_n$. Substitute (4.3) and then $a_1 = 1$. $\square$

Consequently every modulus excluded by the cube-free criterion fails at the very first tested index for the conjectured totient shift. It is not necessary to search far into a period for a mismatch. For the infinite family $m = p^3$,

$$a_{1+p^2(p-1)} \equiv 1 - \chi_4(p)p^2 \pmod{p^3}. \quad (7.3)$$

For $p = 3, 5, 7$, the right sides are respectively 10, 101, 50 modulo 27, 125, 343.

This also shows why replacing the actual least period by a much larger multiple cannot remove the transient: the unit contribution stays fixed, while the nonunit contribution at the early index is still present.

7.3 Two useful local examples

Modulo 3, (4.2) has two equal unit modes. For $n \geq 1$,

$$a_n \equiv 2(-1)^n \pmod{3}. \quad (7.4)$$

Thus $3 \nmid a_n$ for every $n \geq 1$, and

$$s(3^e) = \lceil e/2 \rceil \quad (e \geq 1).$$

The generic upper bound is attained at every power of 3.

At $p = 5, e = 5$, the candidate transient indices are 0, 1, 2. Their values of $2n + v_5(a_n)$ are 0, 2, 5, respectively. Only 0 and 1 are nonzero-transient indices modulo 5^5 , so $s(3125) = 2$, whereas $d(3125) = 1250$. Thus an exact preperiod theorem must retain divisibility information; $s(p^e) = \lceil e/2 \rceil$ is not universally valid.

8 Generating functions and effective computation

8.1 A rational ordinary generating function modulo an odd integer

The ordinary generating series is used formally, without asserting analytic convergence over the complex numbers. Define

$$G_q(z) = \sum_{n \geq 0} (a_n \bmod q) z^n \in (\mathbb{Z}/q\mathbb{Z})[[z]].$$

For odd q , summing (4.2) coefficientwise gives

$$G_q(z) = \sum_{j=0}^{q-1} \frac{(-1)^j}{1 + (2j+1)^2 z}. \quad (8.1)$$

Every denominator has constant coefficient 1, hence is invertible as a formal power series even when q is composite.

At a prime power, denominators with a nonunit ratio contribute only polynomials: if $p \mid 2j + 1$, then the geometric series stops after at most $\lceil e/2 \rceil$ terms because its ratio is nilpotent. Unit ratios produce the periodic part. Thus the polynomial-plus-periodic split of the coefficient sequence is visible directly in the rational series.

For an arbitrary modulus, put $s = s(m)$, $d = d(m)$ and write $\bar{a}_n = a_n \bmod m$. Then

$$G_m(z) = \sum_{n=0}^{s-1} \bar{a}_n z^n + \frac{z^s}{1 - z^d} \sum_{j=0}^{d-1} \bar{a}_{s+j} z^j. \quad (8.2)$$

An empty initial sum is zero. We claim this explicit representation, not an irreducible denominator or a field-like notion of minimal denominator degree over a residue ring with zero divisors.

8.2 Computing periods without enumerating a period

Given a factorization of m , the period is obtained by integer exponentiation and least common multiples. To compute the onset, let H be the largest $\lceil e/2 \rceil$ among odd prime powers dividing m . Generate $a_0, \dots, a_{H-1}$ and inspect the valuations in (2.3). The number of coefficient operations in the binomial recurrence is $O(H^2)$ when binomial coefficients are available; this is an arithmetic-operation estimate, not a unit-cost bit complexity claim for arbitrarily large integers.

The supplied implementation uses trial division for factorization and primality checks. It is intentionally transparent, not a fast factorization package. It is suitable for modest moduli and for prime-power experiments with modest primes. Factoring a huge arbitrary modulus can be the dominant cost and is not hidden in the period formula.

For a huge index n and a modest odd modulus q , (4.2) computes $a_n \bmod q$ using q modular exponentiations, or $O(q \log(n+1))$ modular multiplications with binary exponentiation. Extra working storage apart from the input integers is constant. If $s(q)$ and $d(q)$ are known and $n \geq s(q)$, the index may instead be reduced to

$$s(q) + (n - s(q)) \bmod d(q).$$

These are complementary approaches: the moment formula handles huge indices without ever constructing huge secant numbers, while the onset formula uses only a short initial segment.

8.3 Independent coefficient generation

For testing, the archive also constructs the coefficients through the Entringer triangle, rather than reusing (1.4). Set $T(0, 0) = 1$, and for $r \geq 1$ use

$$T(r, 0) = 0, \quad T(r, k) = T(r, k-1) + T(r-1, r-k) \quad (1 \leq k \leq r).$$

Then $T(2n, 2n) = a_n$. This is a standard alternating-permutation triangle; its row-end construction is also described in the literature on the full up/down numbers [5]. The implementation stores only the current row and the required even-row endpoints. Comparison of two different constructions helps detect indexing, sign, and recurrence implementation errors; it is not a substitute for the proofs above.

9 How often is the original assertion true?

The counterexamples have an exact arithmetic description, so we can count them rather than just exhibit them. More generally, we can determine the natural density of moduli with any prescribed upper bound on their first periodic index.

For a set S of positive integers, its natural density, when it exists, is

$$\text{dens}(S) = \lim_{X \rightarrow \infty} \frac{\#\{m \leq X : m \in S\}}{\lfloor X \rfloor}.$$

For $N \geq 1$ and an odd prime p , define

$$c_p(N) = \min_{n \geq N} (2n + v_p(a_n)). \quad (9.1)$$

This looks like an infinite minimization, but is actually a finite one.

Lemma 9.1 (Finite cutoff computation). *Let $v = v_p(a_N)$. Then*

$$c_p(N) = \min_{N \leq n \leq N + \lfloor v/2 \rfloor} (2n + v_p(a_n)). \quad (9.2)$$

In particular, $c_p(N) \geq 2N$, with equality whenever $p \nmid a_N$.

Proof. The term at $n = N$ gives the upper bound $2N + v$. If $n > N + \lfloor v/2 \rfloor$, then $2n > 2N + v$, so that index cannot minimize the expression. The final assertions follow at once. $\square$

Theorem 9.2 (Distribution of the first periodic index). *For $N \geq 1$, the set $\{m : s(m) \leq N\}$ has natural density*

$$D_N = \prod_{p \text{ odd}} \left(1 - p^{-(c_p(N)+1)}\right). \quad (9.3)$$

Equivalently,

$$D_N = \frac{1}{(1 - 2^{-(2N+1)})\zeta(2N+1)} \prod_{\substack{p|a_N \\ p \text{ odd}}} \frac{1 - p^{-(c_p(N)+1)}}{1 - p^{-(2N+1)}}. \quad (9.4)$$

The correction product is finite. More precisely, for fixed N ,

$$\#\{m \leq X : s(m) \leq N\} = D_N X + O_N(X^{1/(2N+1)}). \quad (9.5)$$

Proof. By the exact transient criterion, $s(m) \leq N$ if and only if, at every odd prime p , all nonunit contributions vanish at every index $n \geq N$. Writing $e = v_p(m)$, this is equivalent to

$$e \leq 2n + v_p(a_n) \quad \text{for every } n \geq N,$$

or $v_p(m) \leq c_p(N)$. There is no restriction at 2. Thus the set is exactly the set of integers avoiding divisibility by each $p^{c_p(N)+1}$ for odd primes p .

Here is a counting proof that also justifies the infinite Euler product. Set $r = 2N + 1$ and let S be the finite set of odd prime divisors of a_N . Outside S , the forbidden exponent is exactly r . For $p \in S$, put $b_p = c_p(N) + 1 \geq r$. Inclusion–exclusion at the exceptional primes, and Möbius inversion elsewhere, give the exact count

$$\sum_{J \subseteq S} (-1)^{|J|} \sum_{\substack{d \geq 1 \\ (d, 2 \prod_{p \in S} p) = 1}} \mu(d) \left\lfloor \frac{X}{d^r C_J} \right\rfloor, \quad C_J = \prod_{p \in J} p^{b_p}.$$

Only $d \leq (X/C_J)^{1/r}$ contribute. Replacing the floor by its argument costs $O_N(X^{1/r})$, since there are that many potentially nonzero terms for each of finitely many J . Extending the main-term sum over d to infinity also costs $O_N(X^{1/r})$, using $\sum_{d>Y} d^{-r} = O_r(Y^{1-r})$.

The coefficient of X is

$$\prod_{p \in S} (1 - p^{-b_p}) \prod_{\substack{p \text{ odd} \\ p \notin S}} (1 - p^{-r}).$$

This proves (9.3) and (9.5). The Euler product for $\zeta(r)$, valid since $r > 1$, gives (9.4). Absolute convergence also follows from $c_p(N) + 1 \geq r \geq 3$. $\square$

Corollary 9.3 (Density of the moduli satisfying the conjecture). *The natural density of moduli for which (1.3) is true is*

$$D_1 = \prod_{p \text{ odd}} (1 - p^{-3}) = \frac{8}{7\zeta(3)} = 0.95075128294937996420 \dots \quad (9.6)$$

The counterexample moduli have density $1 - D_1$, approximately 4.92487%.

Proof. Since $a_1 = 1$, there are no exceptional primes in (9.4); all cutoffs are $c_p(1) = 2$. $\square$

The restriction is that the *odd part* be cube-free; arbitrary powers of 2 are allowed. Accordingly the density is not $1/\zeta(3)$. The factor $(1 - 2^{-3})^{-1} = 8/7$ accounts exactly for the unrestricted power of two.

10 The whole distribution and its asymptotic tail

10.1 The first exceptional correction

At $N = 2$, $a_2 = 5$, so $c_5(2) = 5$ and all other odd-prime cutoffs are 4. Therefore

$$D_2 = \frac{1 - 5^{-6}}{1 - 5^{-5}} \frac{1}{(1 - 2^{-5})\zeta(5)}. \quad (10.1)$$

The corresponding moduli have odd prime exponents at most 4, except that exponent 5 is allowed at the prime 5. This is the density-level reflection of $s(5^5) = 2$.

At $N = 10$, one must not replace every exceptional cutoff by $2N + 1$:

$$a_{10} = 5^2 \cdot 41737 \cdot 354957173.$$

Here $c_5(10) = 22$, while the other two cutoffs are 21. The finite minimization in Lemma 9.1 captures such higher-valuation effects automatically.

Table 2: Densities $D_N = \text{dens}\{m : s(m) \leq N\}$, rounded to 18 decimal places. Full rational enclosures are supplied in the archive.

N	D_N	N	D_N
1	0.950751282949379964	6	0.999999372600335288
2	0.995751538184580342	7	0.99999930275301732
3	0.999528673601264576	8	0.99999992256209180
4	0.999949067031528573	9	0.99999999139555886
5	0.99999433981021976	10	0.99999999904400848

Only powers of two have $s(m) = 0$, and their density is zero. Define $D_0 = 0$. For every $N \geq 1$, the moduli with exact first periodic index N have density $D_N - D_{N-1}$. Thus the cumulative densities determine an entire probability distribution in the limit of uniformly sampled moduli.

10.2 The tail is controlled by the prime three

Theorem 10.1 (Asymptotic distribution tail). *As $N \rightarrow \infty$,*

$$1 - D_N = 3^{-(2N+1)} + O\left(5^{-(2N+1)}\right). \quad (10.2)$$

In particular, the limiting distribution has finite moments of every positive integer order.

Proof. Equation (7.4) shows that $c_3(N) = 2N$ for every $N \geq 1$. Thus the set of multiples of 3^{2N+1} is included in the excluded moduli and has density $3^{-(2N+1)}$. Every other prime has forbidden exponent at least $2N + 1$. A union bound gives, with $r = 2N + 1$,

$$3^{-r} \leq 1 - D_N \leq 3^{-r} + \sum_{p \geq 5} p^{-r}.$$

For $r \geq 3$, the last sum is at most

$$\sum_{k=5}^{\infty} k^{-r} \leq 5^{-r} + \frac{5^{1-r}}{r-1} = O(5^{-r}),$$

with an absolute constant. This proves the expansion. Exponential decay of the tail implies finiteness of every polynomial moment by the usual discrete tail-sum formula. $\square$

The theorem has a concrete interpretation: among very long transients, the overwhelmingly dominant cause is a high power of 3 in the modulus. Exceptional divisibility of secant numbers can only shorten transients; it cannot create a forbidden exponent below $2N + 1$ at another prime.

11 The mean first periodic index

Theorem 11.1 (Mean and a certified enclosure). *The arithmetic mean of $s(m)$ over initial intervals has a limit:*

$$\mathcal{M} = \lim_{X \rightarrow \infty} \frac{1}{\lfloor X \rfloor} \sum_{m \leq X} s(m) = 1 + \sum_{N \geq 1} (1 - D_N). \quad (11.1)$$

It satisfies the rigorous bounds

$$1.05402581008837151 < \mathcal{M} < 1.05402581008837162. \quad (11.2)$$

Proof of the limit formula. For every nonnegative integer-valued function,

$$s(m) = \sum_{N \geq 0} \mathbf{1}_{\{s(m) > N\}}.$$

For $N = 0$, the limiting proportion is 1, since $s(m) = 0$ only at powers of two. For fixed $N \geq 1$, the limiting proportion is $1 - D_N$. We must justify passing the limit through the infinite sum.

If $s(m) > N$, then some odd prime power $p^e \parallel m$ satisfies $e > c_p(N) \geq 2N$, so $p^{2N+1} \mid m$. Consequently, for any positive integer M ,

$$\frac{\#\{m \leq M : s(m) > N\}}{M} \leq \sum_{p \text{ odd}} p^{-(2N+1)}.$$

The sum of these bounds over $N \geq 1$ converges. Dominated convergence for the counting averages now proves (11.1). The enclosure follows from the rational computations and tail estimates below, whose numerical method is justified in Appendix B. $\square$

Let

$$\mathcal{M}_K = 1 + \sum_{N=1}^K (1 - D_N).$$

The proof of Theorem 10.1 gives a particularly simple certified tail:

$$\frac{9}{8} 3^{-(2K+3)} \leq \mathcal{M} - \mathcal{M}_K \leq \frac{9}{8} \left((1 - 2^{-(2K+3)}) \zeta(2K+3) - 1 \right). \quad (11.3)$$

For the lower bound, sum the mandatory prime-3 contributions. For the upper bound, sum the union bound over primes and use $(1 - p^{-2})^{-1} \leq 9/8$ for every odd prime. Enlarging the sum from odd primes to all odd integers at least 3 gives exactly the displayed zeta expression.

Taking $K = 10$ and evaluating the finitely many zeta factors with rational interval arithmetic yields

$$\begin{aligned} 1.054025810088371518561523366670 &< \mathcal{M}, \\ \mathcal{M} &< 1.054025810088371612974596863433. \end{aligned}$$

The slightly wider decimal bounds in (11.2) are convenient to quote. They certify the rounded value $\mathcal{M} \approx 1.05402581008837$ without relying on floating-point rounding or an unproved estimate of a zeta-series remainder.

12 Computational verification and research scope

12.1 What was actually run

The accompanying Python programs use only the standard library. The verification run used Python 3.13.5; the source is written for Python 3.9 or later. The exact range checks recorded in `data/verification.json` are summarized in Table 3. Ordinary integer and rational arithmetic is used throughout; the density enclosures do not use floating-point values.

For each modulus through 1000, the code checks a complete proposed period block and its shift, verifies a mismatch at $s(m) - 1$ when the preperiod is nonzero, and rejects each candidate $d(m)/\ell$ with ℓ a prime divisor of $d(m)$. These witnesses are stored explicitly. Why are those candidates enough? Any proper divisor of $d(m)$ divides at least one such $d(m)/\ell$, and a sequence with the smaller period would also have the larger multiple as a period.

These finite checks support the implementation and catch mistakes, but do not prove that a pattern observed on a finite prefix persists forever. The universal statements rest on the mathematical proofs in the report. No proof-assistant formalization was executed. The CSV entries for $1001 \leq m \leq 10000$ are outputs of the proved formulas, not a claim of independent coefficient-by-coefficient testing for that extra range.

The separate density program uses 96 terms of a positive Euler-transformed eta series for each required zeta value. Its JSON file records exact rational endpoints, outward-rounded

Table 3: Completed checks. Formula-derived data beyond the independently checked range are distinguished explicitly.

Check	Count or range
Exact secant numbers generated by the triangle	$0 \leq n \leq 2100$
Agreement with the independent binomial recurrence	$0 \leq n \leq 100$
Finite-moment evaluations against exact coefficients	1900
Moduli checked for period and onset consistency	$1 \leq m \leq 1000$
Equalities at the proposed period shift	130793
Proper-divisor period candidates rejected by witnesses	2432
All-index Euler-factor defect checks	651
Huge-index reductions, including indices above 10^{100}	18
Arithmetic checks of the two lifting-mode coefficients	15
Moduli in the formula-generated CSV table	$1 \leq m \leq 10000$
Densities supplied with exact rational interval certificates	$1 \leq N \leq 10$

decimal endpoints, the exceptional prime factors, and the finite cutoff calculations. A reader can rerun every reported finite calculation without network access or a computer algebra system.

12.2 What is settled, and what remains a different problem

The OEIS assertion (1.3) is false, and its correct domain of validity is exactly characterized. There is no remaining conjectural step in the stated local periods, finite onset formula, global assembly, density formulas, or mean formula. The proofs are conventional arguments and are open to independent checking; the computations are supplementary.

A different question is whether the valuation tests in (2.3) admit a dramatically simpler closed form, or how large the gap $\lceil e/2 \rceil - s(p^e)$ can become. The complete effective criterion does not by itself answer those structural questions. Likewise, the remaining conjectures about the *full* up/down sequence in [6] are not resolved here. Confusing that sequence with its even subsequence would incorrectly turn a specialized result into a claim about a different open problem.

The useful methodological point is narrow and concrete. A modular sequence represented by finitely many power functions naturally divides into unit modes and nilpotent modes. Pure periodicity can fail precisely because the latter survive at a few initial indices. Identifying that transient exactly converts an isolated counterexample into a complete classification, and then into an arithmetic distribution problem.

A A standalone counterexample checker

The following complete program requires Python 3.8 or later. It checks only the finite disproof and does not depend on any theorem in the report. The full archive uses Python 3.9 or later because it also uses `math.lcm`.

```
from math import comb

# sec(x) * cos(x) = 1, with integer coefficient arithmetic.
a = [1]
for n in range(1, 20):
    value = sum(
        (1 if j % 2 else -1) * comb(2*n, 2*j) * a[n-j]
        for j in range(1, n+1)
    )
    a.append(value)

print("a(19) =", a[19])
print("a(1) mod 27 =", a[1] % 27)
print("a(19) mod 27 =", a[19] % 27)

# phi(27) = 18. A period dividing 18 forces these equal.
if a[1] % 27 == a[19] % 27:
    raise RuntimeError("The claimed counterexample failed")
print("Conjecture disproved: 1 != 10 modulo 27.")
```

Output:

```
a(19) = 23489580527043108252017828576198947741
a(1) mod 27 = 1
a(19) mod 27 = 10
Conjecture disproved: 1 != 10 modulo 27.
```

The same recurrence may be reduced modulo 27 at every step, avoiding large integers entirely. We retain the exact value here so that the counterexample can be checked in either form.

B Rational certificates for the numerical constants

For an integer $r > 1$, let

$$\eta(r) = \sum_{j \geq 0} \frac{(-1)^j}{(j+1)^r} = (1 - 2^{1-r})\zeta(r).$$

Define positive finite differences

$$\Delta_{k,r} = \sum_{j=0}^k (-1)^j \binom{k}{j} (j+1)^{-r}.$$

The elementary integral identity

$$\Delta_{k,r} = \frac{1}{(r-1)!} \int_0^1 (-\log t)^{r-1} (1-t)^k dt \quad (\text{B.1})$$

shows $0 < \Delta_{k,r} \leq 1$. Summing the geometric series in the integral also proves the Euler transform

$$\eta(r) = \sum_{k \geq 0} \frac{\Delta_{k,r}}{2^{k+1}}. \quad (\text{B.2})$$

Indeed, the sum of $(1-t)^k/2^{k+1}$ is $1/(1+t)$; integrating its geometric expansion in t gives the eta series. Positivity or dominated convergence justifies the exchanges.

If

$$H_{K,r} = \sum_{k=0}^{K-1} \frac{\Delta_{k,r}}{2^{k+1}},$$

then both endpoints of

$$\frac{H_{K,r}}{1 - 2^{1-r}} \leq \zeta(r) \leq \frac{H_{K,r} + 2^{-K}}{1 - 2^{1-r}} \quad (\text{B.3})$$

are rational, and the enclosure follows because the omitted positive terms sum to at most 2^{-K} . The program forms the finite differences recursively with `Fraction`, not by cancellation-prone floating-point subtraction.

For D_N , the finite correction factor in (9.4) is an exact rational number. The reciprocal of the upper zeta endpoint gives a lower density endpoint, and the reciprocal of the lower zeta endpoint gives an upper density endpoint. Summing those bounds with the direction reversed for $1 - D_N$, then applying (11.3), gives the mean certificate. Decimal conversions are rounded outwards using the Python `decimal` module.

The certificate file includes unusually long integers because it stores exact common denominators rather than a rounded approximation. On Python versions with a decimal-digit limit for integer serialization, the certificate program disables that limit only for these locally generated integers. It does not parse untrusted large integer strings or fetch external input.

C Archive contents and reproduction

The principal files are `article.tex`, `article.pdf`, `secant_periodicity.py`, `verify.py`, and `certify_densities.py`. The `data/` directory contains all tables, witnesses, logs, and rational certificates. A separate `counterexample.py` reproduces Appendix A. The file `oeis_correction.txt` is a proposed correction note for human review; nothing has been submitted to the OEIS.

From the archive directory, run

```
python counterexample.py
python verify.py
python certify_densities.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first three commands need no third-party Python packages. Building the PDF requires a LaTeX installation with the packages listed in the source, including New PX text and math fonts. Font files are not included in the archive. A Makefile, a PowerShell build script, and a source-provenance note are included for convenience.

References

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